Model	*Venue*	*Training Type*	*Datasets*	*Github Link*	*Schema-linking results*
DAIL-SQL	None (2023)	Prompting/ICL	Spider, BIRD	https://github.com/beachwang/dail-sql	
C3	None (2023)	Prompting/ICL	Spider	https://github.com/bigbigwatermalon/C3SQL	
DIN-SQL	NeurIPS 2023	Prompting/ICL	Spider, BIRD	https://github.com/MohammadrezaPourreza/Few-shot-NL2SQL-with-prompting	
PET-SQL	None (2024)	Prompting/ICL	Spider	https://github.com/zhshLii/PETSQL	Yes
RSL-SQL	None (2024)	Prompting/ICL	Spider, BIRD	https://github.com/Laqcce-cao/RSL-SQL	Yes
OmniSQL	None (2025)	Supervised Fine-tuning	Spider 2.0-Lite, Spider, BIRD	https://github.com/RUCKBReasoning/OmniSQL	
MAC-SQL	COLING 2025	Prompting/ICL	Spider, BIRD	https://github.com/wbbeyourself/MAC-SQL	
MCS-SQL	COLING 2025	Prompting/ICL	Spider, BIRD		
CHESS	ICML 2025	Prompting/ICL	Spider, BIRD	https://github.com/shayantalaei/chess	
CHASE-SQL	ICLR 2025	Prompting/ICL	Spider, BIRD	https://github.com/kungfuai/chase-sql	
Alpha-SQL	ICML 2025	Prompting/ICL	Spider, BIRD	https://github.com/HKUSTDial/Alpha-SQL	
Reasoning-SQL	COLM 2025	Supervised Fine-tuning Reinforcement Learning	Spider, BIRD		
Dialect-SQL	EMNLP 2025	Prompting/ICL	Spider, BIRD	https://github.com/jieshi10/orm-sql	
LinkAlign	EMNLP 2025	Prompting/ICL	Spider 2.0-Lite, Spider, BIRD	https://github.com/Satissss/LinkAlign	Yes
Safe-SQL	EMNLP 2025	Prompting/ICL	Spider, BIRD		
JOLT-SQL	EMNLP 2025	Supervised Fine-tuning	Spider, BIRD	https://github.com/Songjw133/JOLT-SQL	Yes
GenLink	EMNLP 2025	Supervised Fine-tuning	Spider, BIRD	https://github.com/DMIRLAB-Group/GenLink	Yes
AutoLink	AAAI 2026	Prompting/ICL	Spider 2.0-Lite, BIRD	https://github.com/wzy416/AutoLink	Yes
MCTS SQL	AAAI 2026	Prompting/ICL	Spider, BIRD		
Schemalinking	EACL 2026	Prompting/ICL	Spider, BIRD	https://github.com/mahadi-nahid/BiSchemaLink	Yes
SchemaGraphSQL	EACL 2026	Prompting/ICL	BIRD, Spider 2.0 (Lite)		Yes
LiteSQL	EACL 2026	Supervised Fine-tuning Reinforcement Learning	Spider, BIRD	https://github.com/shengminp/LitE-SQL	Yes
AST-SQL	ICML 2026	Reinforcement Learning	Spider, BIRD, Spider 2.0 (Appendix)		
SquRL	ICML 2026	Reinforcement Learning	Spider 2.0-Lite, Spider, BIRD	https://github.com/Satissss/SquRL	
Graph-Link	ICML 2026	Prompting/ICL	Spider 2.0-Lite, BIRD		Yes
MARS-SQL	ICML 2026	Supervised Fine-tuning Reinforcement Learning	Spider, BIRD		

Model	*Venue*	*Training Type*	*Datasets*	*Github Link*	*Schema-linking results*
DPC	ACL2026	Prompting/ICL	Spider, BIRD	https://github.com/HKUSTDial/DPC	
GBV-SQL	ACL2026	Prompting/ICL	Spider, BIRD		
PV-SQL	ACL2026	Prompting/ICL	Spider, BIRD	https://github.com/magic-YuanTian/PV-SQL	
ReEx	ACL2026	Reinforcement Learning	Spider, BIRD		
SQL-Trail	ACL2026	Supervised Fine-tuning Reinforcement Learning	Spider, BIRD		
SDE-SQL	ACL2026	Prompting/ICL Supervised Fine-tuning	Spider, BIRD	https://github.com/Lancelot-Xie/SDE-SQL	
Arctic-Text2SQL	ACL2026	Reinforcement Learning	Spider, BIRD	https://github.com/snowflakedb/ArcticTraining/	
VET	ACL2026	Prompting/ICL	Spider 2.0-Lite, Spider, BIRD		
EXPO-SQL	ACL2026	Reinforcement Learning	Spider, BIRD	https://github.com/jhn25/EXPO-SQL	
R3-SQL	ACL2026	Prompting/ICL	Spider, BIRD	https://github.com/snowflakedb/ArcticTraining/tree/main/projects/arctic_text2sql_r1	
PExA	ACL2026	Prompting/ICL	Spider 2.0-Lite		
XiYan-SQL	IEEE TKDE 2026	Supervised Fine-tuning	Spider, BIRD	https://github.com/XGenerationLab/XiYan-SQL	

Model	*Venue (Year)*	*Backbone*	*EX (%)*		*Precision*	*Recall*

Method	*Venue (Year)*	*Embedding Space*	*Backbone (LLM)*	*EX (%) - Total*	*EX (%) - Simple*	*EX (%) - Moderate*	*EX (%) - Challenging*	*Tokens*		*Precision*	*Recall*	*Hit Rate*
Zero-Shot			Qwen3-max (95B)									
Full Schema (No Link)				45.7	52.4	40.3	21.5	~ 55,300				
Hybrid (BM25 + Vector)				29.4	38.4	18.6	6.9	~ 8,500				
DIN-SQL (Linker-Only)		Text-embedding-v4	Qwen3-max (95B)/GLM4.7 (32B)	52.4	58.3	48.1	28.9	~ 33,200				
MAC-SQL (Linker-Only)		Text-embedding-v4	Qwen3-max (95B)/GLM4.7 (32B)	54.8	59.6	52.3	32.9	~ 15,800				
MCS-SQL (Linker-Only)		Text-embedding-v4	Qwen3-max (95B)/GLM4.7 (32B)	66	72.8	59.3	44.8	~ 63.800				
RSL-SQL (Linker-Only)		Text-embedding-v4	Qwen3-max (95B)/GLM4.7 (32B)	63.4	69.6	58.4	40.8	~ 48,100				
Graph-Link		Text-embedding-v4	Qwen3-max (95B)/GLM4.7 (32B)	68.5	73.3	62.2	58.6	~ 20,800				
SlayQL												

Model	***Venue (Year)***	***Backbone***	***EX (%)***		***Precision***	***Recall***
LinkAlign (DeepSeek)	EMNLP (2025)		33.09			
AutoLink (DeepSeek)	AAAI (2026)		34.92			91.20
SchemaGraphSQL	EACL (2026)				59.88	47.37
AST-SQL (Qwen-2.5-Coder)	ICML (2026)		8.90			
AST-SQL (Omini-SQL-7B)	ICML (2026)		14.10			
Graph-Link	ICML (2026)				69.7	91.30
SquRL	ICML (2026)		***49.18***			
Arctic-Text2SQL-R1-32B	ACL (2026)		16.30			
VET	ACL (2026)	DeepSeek-Chat	37.04			
PExA	ACL (2026)		46.60			
SlayQL			45.89			

Model	***Venue (Year)***	***Backbone***	***EX (%)***		***Precision***	***Recall***
PET-SQL	None (2004)		87.60			86.40
Dialect-SQL (DeepSeek)	EMNLP (2025)		83.00			
GenLink	EMNLP (2025)		87.80			
Safe-SQL (Gemma3-27B)	EMNLP (2025)		88.50			
JOLT-SQL (Qwen2.5-Coder-14B)	EMNLP (2025)		88.90		93.74	98.58
LinkAlign (DeepSeek)	EMNLP (2025)		88.90			87.30
MCTS SQL (Gemini)	AAAI (2026)		89.17			
Schemalinking (Gemini)	EACL (2026)		70.60			
LiteSQL	EACL (2026)		88.45			99.16
AST-SQL	ICML (2026)		83.00			
SquRL	ICML (2026)		86.58			
MARS-SQL	ICML (2026)		***89.75***			
DPC (DeepSeek)	ACL (2026)		73.20			
PV-SQL + GPT-4o	ACL (2026)		77.66			
GBV-SQL (DeepSeek v3)	ACL (2026)		79.60			
VET	ACL (2026)	GPT-4o	84.90			
SQL-Trail-14B (Majority)	ACL (2026)		87.10			
R3-SQL	ACL (2026)		87.19			
SDE-SQL	ACL (2026)	Qwen2.5-72B-Instruct	87.30			
Arctic-Text2SQL-R1-32B	ACL (2026)		88.70			
ReEX-SQL	ACL (2026)		89.10			
EXPO-SQL	ACL (2026)	Qwen2.5-Coder-14B	89.20			
XiYan-SQL	IEEE TKDE (2026)	Qwen2.5-Coder-32B	89.65			

Model	*Venue (Year)*	*Backbone*	*EX (%)*		*Precision*	*Recall*
RSL-SQL	None (2024)	GPT-4o	67.21			94.32
MCS-SQL	COLING 2025	GPT-4	65.50			99.20
CHESS	ICML 2025	Gemini 1.5-pro	68.31			
Dialect-SQL (DeepSeek)	EMNLP (2025)	DeepSeek-R1-Distill-Qwen-32B	56.84			
LinkAlign (DeepSeek)	EMNLP (2025)	GPT-4	61.60			79.40
Safe-SQL (Gemma3-27B)	EMNLP (2025)	GPT-4o	63.50			
JOLT-SQL (Qwen2.5-Coder-14B)	EMNLP (2025)	Qwen2.5-Coder-14B	64.90		75.12	96.68
GenLink	EMNLP (2025)	Llama-3.1-8BInstruct Qwen2.5-Coder-7B-Instruct Qwen2.57B-Instruct Deepseek-coder-6.7b-instruct Mistral-7B-Instruct-v0.3	67.34			94.56-97.22 [1]
AutoLink	AAAI (2026)	DeepSeek	68.71			97.40
MCTS SQL (Gemini)	AAAI (2026)		72.91			
Schemalinking	EACL (2026)	Gemini 2.0-Flash	57.80		-	90.60
SchemaGraphSQL	EACL (2026)	Gemini 2.5-Flash	62.91		86.21	95.71
LiteSQL	EACL (2026)	Qwen2.5-Coder	72.10			82.31
AST-SQL	ICML (2026)	Qwen2.5-Coder-7B	65.78			
SquRL	ICML (2026)		67.60			
Graph-Link	ICML (2026)	Qwen3-max	68.50		70.40	95.90
MARS-SQL	ICML (2026)		***77.84***			
DPC (DeepSeek)	ACL (2026)		53.40			
GBV-SQL (DeepSeek v3)	ACL (2026)		63.23			
PV-SQL + GPT-4o	ACL (2026)		65.12			
ReEx	ACL (2026)		65.30			
SQL-Trail-14B (Majority)	ACL (2026)		66.70			
SDE-SQL	ACL (2026)	Qwen2.5-72B-Instruct	68.19			
Arctic-Text2SQL-R1-32B	ACL (2026)		70.50			
VET	ACL (2026)	GPT-4o	70.93			
EXPO-SQL	ACL (2026)	Qwen2.5-Coder-14B	73.00			
R3-SQL	ACL (2026)		75.03			
XiYan-SQL	IEEE TKDE (2026)	Qwen2.5-Coder-32B	75.63			

[1] Count-based Table Recall, Count-based Column Recall